Design Note - Sircles Inbound Lead Agent Team

The four business questions

1. Cost of being wrong: false "pursue" vs. false "skip". The two errors are not symmetric, and they are not even the same *kind* of cost. A false **skip** is an opportunity cost - a lost retainer - but it is **recoverable**: the lead sits in a nurture queue and can re-enter later. A false **pursue** has a cheap part (SDR/strategist time) and an expensive, **irreversible** part: a clumsy first-touch to the wrong or sensitive prospect can "burn them permanently." So the dominant risk is not *chasing* a weak lead, it is *sending* bad outreach autonomously. That single insight shapes the whole disposition logic: skipping is cheap and reversible, so we skip freely; sending is irreversible, so we gate it hard. Under genuine uncertainty we default to **nurture** - the low-regret middle that neither burns the prospect nor discards the revenue.

2. What makes a lead worth it: ICP + signals -> pursue / nurture / skip. Fit is a deterministic, auditable 0-100 score (not an LLM guess): decision-maker seniority (20), company size/budget fit (20), industry fit (20), **location fit - Egypt-weighted (15)**, growth/intent signals (15), and marketing-maturity gap (10). Location is explicit because Sircles operates from Egypt: same timezone, language, payment rails, and local-market knowledge make Egyptian leads the core ICP, MENA gets partial credit, and far-geo is deprioritised but not disqualified. Bands: `>=70 pursue`, `40-69 nurture`, `<40 skip`. Three **hard disqualifiers** override the score: a competing agency (-> escalate, not auto-skip - could be a partnership), a job-seeker/student, and a no-budget micro-entity. A "no budget yet" signal caps an otherwise strong lead at nurture. I separated **fit** (the score) from **confidence** (driven by data completeness) so thin data lowers our willingness to act without changing our read of the opportunity.

3. Confidence -> action: act autonomously vs. escalate. The team acts with no human in the loop only on the happy path: disposition `pursue` **and** confidence `>= 0.70` **and** no red flags -> draft and "send." A `pursue` below 0.70 is drafted but held for human approval (we never auto-send when unsure). `nurture` / `skip` are low-stakes and run autonomously. We **escalate to a human** only when it is genuinely warranted: (a) the critic vetoes (competitor), (b) after a bounded rebuttal the agents remain split on a *polar* call (pursue vs skip) with comparable confidence, or (c) data completeness is below 0.5. Crucially, an *adjacent* disagreement (pursue vs nurture) does **not** escalate - the ICP band breaks that tie - so we never burden a human with an obvious lead. The threshold is a direct expression of the cost asymmetry: the irreversible action requires earned confidence.

4. Where it earns its keep + what to automate next. The money is not in drafting emails - it is in **triage at the top of the funnel**: confidently and consistently skipping/escalating the ~50-70% of inbound that isn't worth a human's time, so SDRs spend their hours only on real opportunities, and never send a reputation-damaging first-touch. That is where the agency saves money (team time) and protects revenue (prospect goodwill). Next I'd automate **reply handling and meeting booking** (the next irreversible touch-points), then the **Onboarding** and **Marketing Strategy** agents the brief hints at.

Tradeoffs I made

- **Deterministic Arbiter + scoring, LLM only for judgment/prose.** The most consequential logic (resolution rule, ICP) is plain Python so it is testable and defensible; LLMs argue and write, they don't silently decide. Trade-off: less "magic," but every decision is auditable.
- **Advocate/Skeptic split** rather than a voting panel of N agents. Genuinely opposed objectives guarantee real disagreement on hard cases without the cost/latency of a crowd. Trade-off: two viewpoints can both miss a third framing.
- **Bounded rebuttal (max 1 round)** then escalate. Prevents infinite or performative debate; trade-off: occasionally escalates something a third round might have resolved - acceptable, because escalation is cheap and a wrong autonomous send is not.
- **Competitor -> escalate, not skip.** Slightly more human load, but a competing agency can be a white-label partner; auto-skipping would throw that away.
- **Everything in one notebook** for reviewability over production packaging.

What I'd do with two more weeks

- Replace the fixture tools with real Apollo/Clay/Hunter + HubSpot adapters behind the same typed interfaces.
- Calibrate thresholds on historical won/lost data instead of hand-set weights; learn the ICP weights.
- Add an LLM-as-judge eval harness + golden traces to catch debate/regression drift across model versions.
- Persist traces to a store and extend the included live Streamlit UI (`app.py`, which already streams the debate and resolution in real time) into a reviewer console for the escalation queue with one-click approve/deny that feeds back into calibration.
- Add guardrails on outreach (PII, tone, claims grounding) before any real "send."

Where it breaks at scale

- **LLM variance & cost:** at thousands of leads/day, two-agent debate on every lead is expensive and non-deterministic. Fix: only invoke the full debate near the decision boundary; auto-decide the obvious top/bottom by score.
- **Stale/ wrong enrichment** silently corrupts every downstream step; the score is only as good as the data. Needs freshness checks and confidence penalties on stale fields.
- **Hand-tuned weights** won't generalise across segments/regions; they need to be learned and monitored for drift.
- **Single shared-state, in-process orchestrator** is fine for a notebook but needs a durable queue, retries, idempotent CRM upserts, and rate-limit handling to run as a real pipeline.
- **Escalation queue overflow:** if too much escalates, humans become the bottleneck - the thresholds must be tuned against available reviewer capacity.